TrES-3b

R-0096

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_250325` · created 2026-07-15T23:08:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460759.9495 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.022
Transit depth (Rp/Rs)^2	2.29 +/- 0.66 [%]
Orbital Inclination (inc)	83.01 +/- 0.88
Airmass coefficient 1 (a1)	0.986 +/- 0.013
Airmass coefficient 2 (a2)	0.012 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.34 %
Best Comparison Star	#2 - [294, 271]
Optimal Aperture	3.51
Optimal Annulus	11.38
Transit Duration (day)	0.0624 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.022 vs 0.16309 (deviation 0.55σ)
Duration fit vs published	0.0624 d vs 0.0567 d (deviation 0.82σ)
Mid-time O–C	3.2 min
Depth SNR	6.9
Residual scatter	1.34 %

Light curve

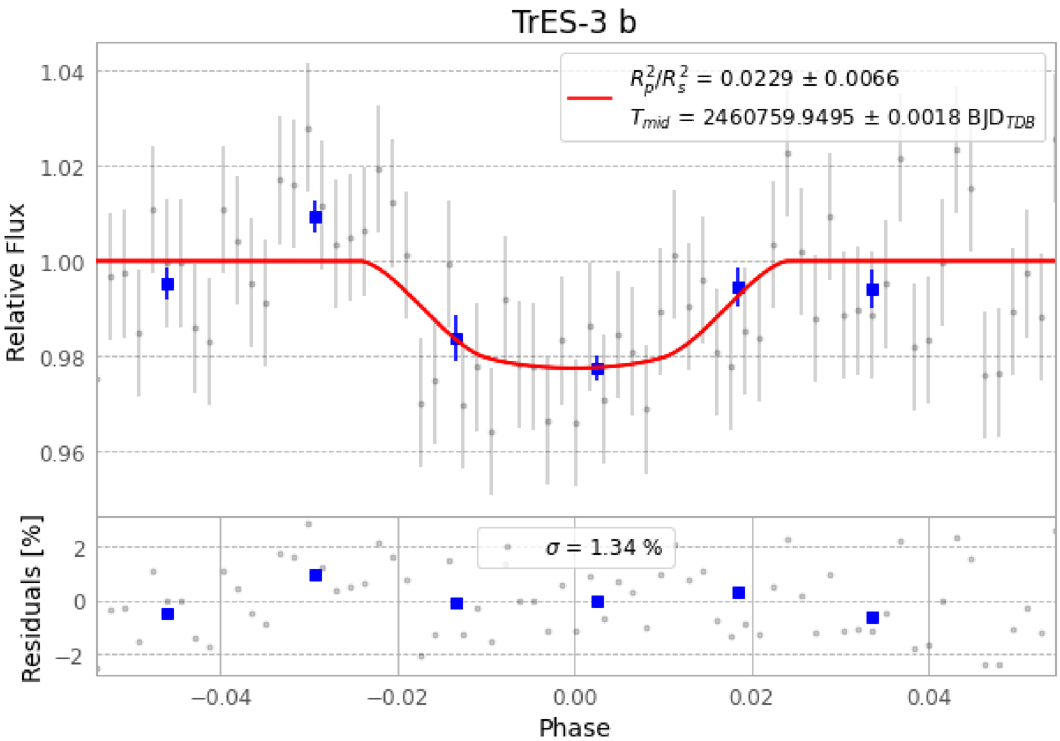

Figure 1 — FinalLightCurve_TrES-3 b_2025-03-25.png (SHA-256 47c03ea13657bae0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 187], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fa7f2d7e69dcb627…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

69 FITS frames (45 MB), `TRES-3250325090316.FITS` → `TRES-3250325122716.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250325.log (42488155d983…), FinalLightCurve_TrES-3 b_2025-03-25.png (47c03ea13657…), FinalLightCurve_TrES-3 b_2025-03-25.csv (458951aa0dd1…)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)